# CONDUCTING SYSTEM OF THE HEART, BLOOD SUPPLY AND NERVE SUPPLY TO HEART

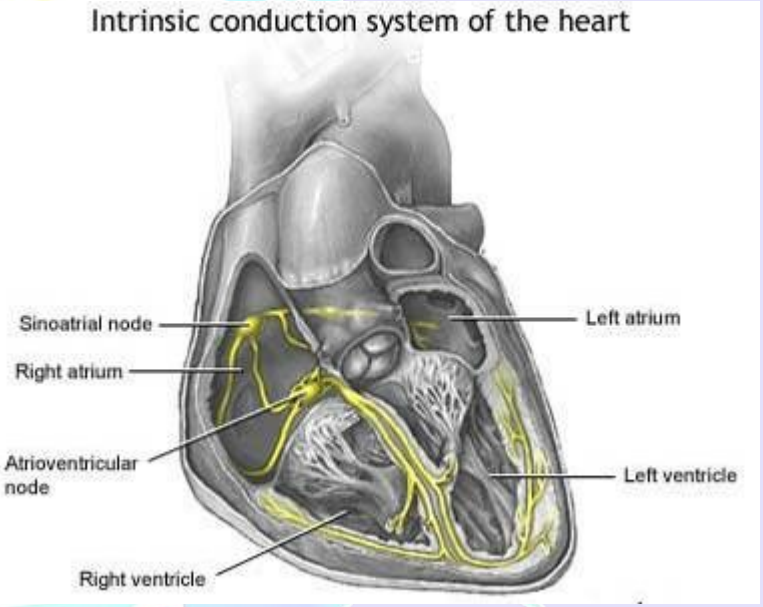
# LEARNING OBJECTIVES

At the end of the lecture the student should be able to:

- Describe the conduction system of heart
- Express different components of conduction system
- Describe the blood supply of heart
- Explain coronary arteries and its branches
- Discuss about the innervation of heart

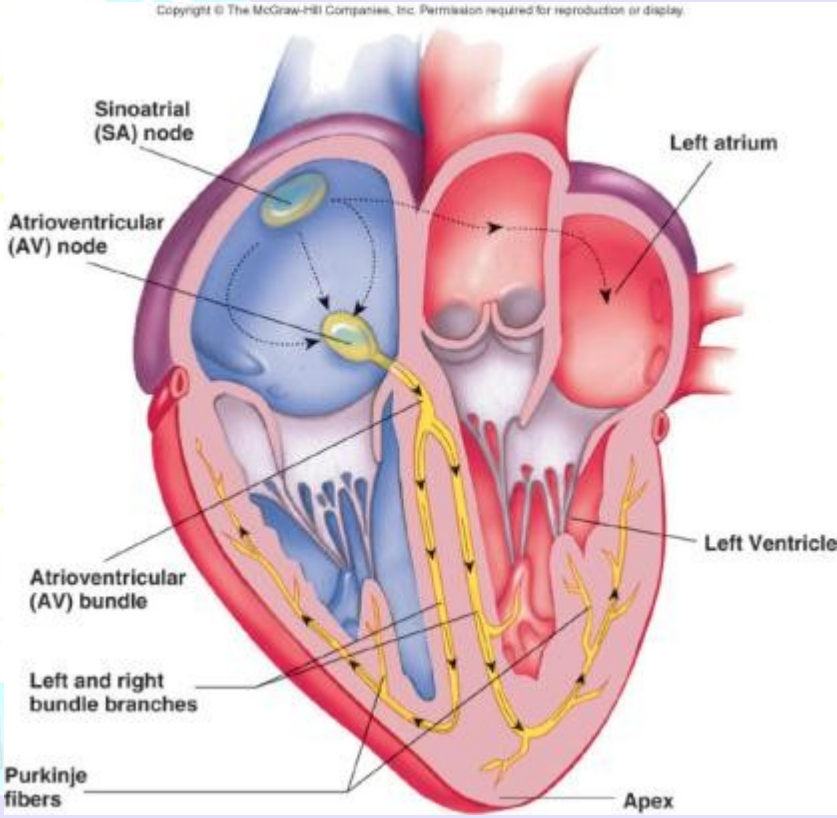
# CONDUCTING SYSTEM OF HEART

- Consists of specialized cardiac muscles present in the
- **Sinuatrial node**, the
- **Atrioventricular node**,
- The **atrioventricular bundle** and its right and left terminal branches,
- And the subendocardial plexus of **purkinje fibers**



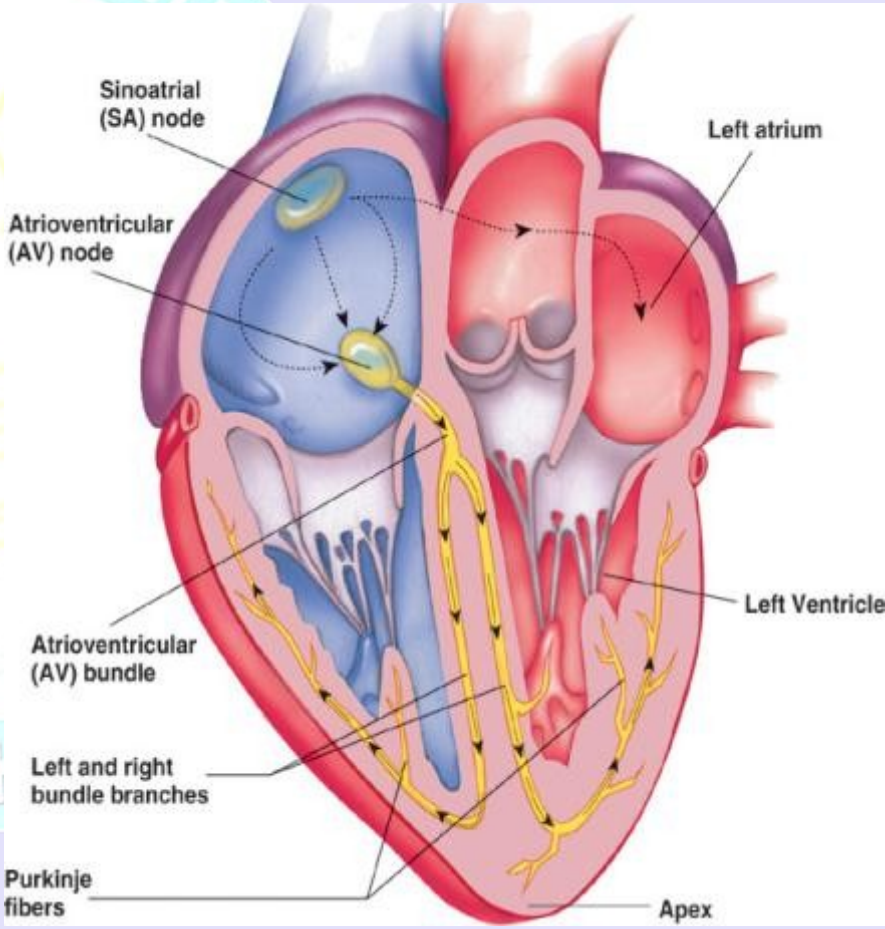
# CONDUCTING SYSTEM OF HEART

- **Conductivity**: ability to transmit an electrical impulse from one cell to another
- Normal heart contracts **rhythmically** at about 70 to 90 beats per minute in the resting adult.
- The impulse travels to different regions of the heart
- The **atria** contract **first** and together
- Followed **later** by the contractions of both **ventricles together**
- Slight delay in the passage of the impulse from the atria to the ventricles allows time for the atria to empty their blood into the ventricles before the contraction of ventricles



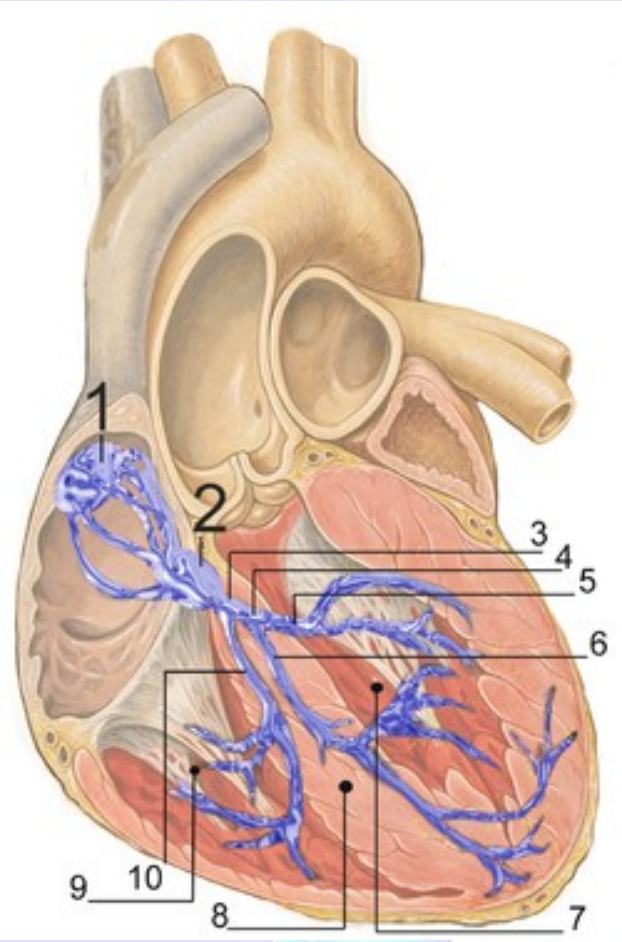
# SINUATRIAL NODE

- Located in the wall of the **right atrium**
- In the upper part of the **sulcus terminalis**
- To the right of the opening of the **superior vena cava**
- Spontaneously gives origin to **rhythmic electrical impulses**
- Spread in all directions through the cardiac muscle of the atria and cause the muscles to contract

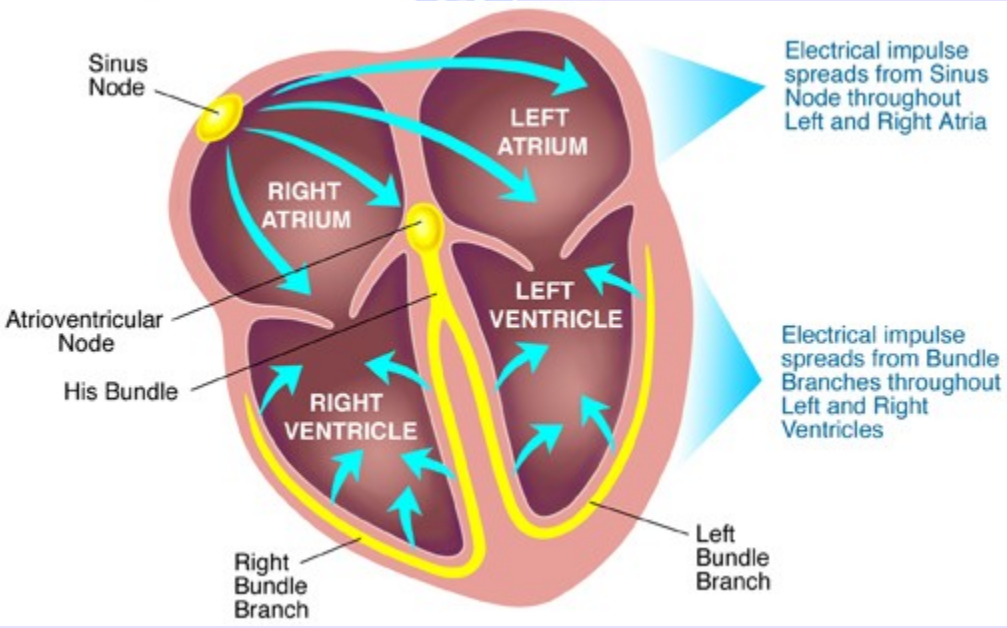


# ATRIOVENTRICULAR NODE

- Placed on the lower part of the atrial septum
- Just above the attachment of the septal cusp of the tricuspid valve.
- From it, the cardiac impulse is conducted to the ventricles by the atrioventricular bundle
- There is a delay of about 0.12 sec in conduction through AV node
- Allow atria to empty blood completely into the ventricles




1. Sinoatrial node
2. Atrioventricular node
3. Bundle of His
4. Left bundle branch
5. left anterior fascicle
6. left posterior fascicle
7. Left ventricle
8. Ventricular septum
9. Right ventricle
10. Right bundle branc

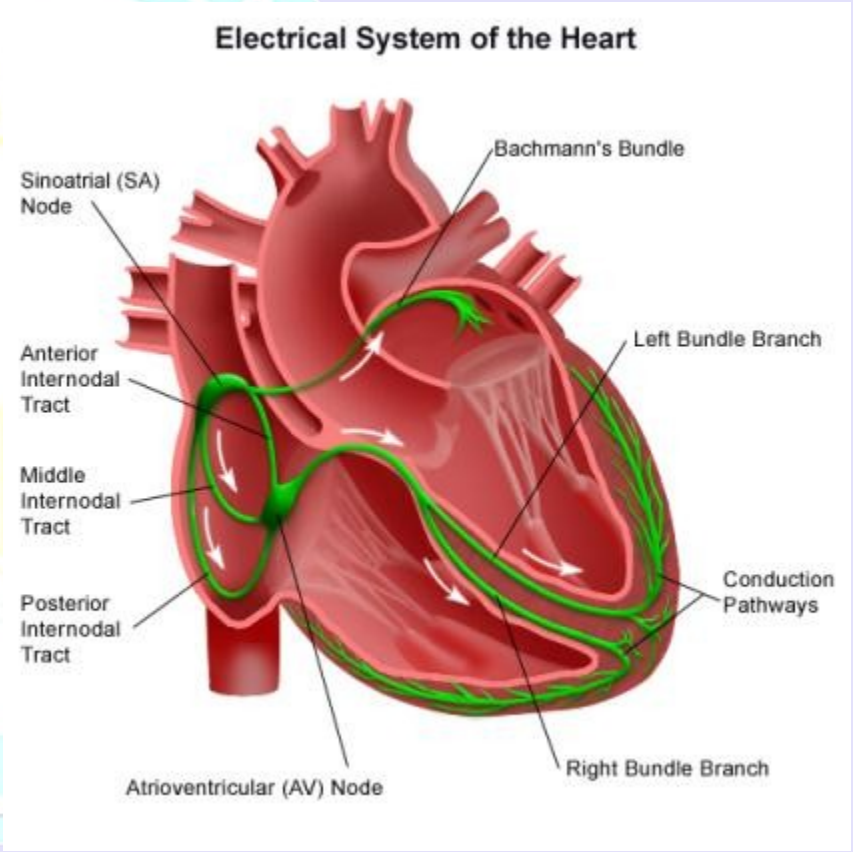


# INTERNODAL CONDUCTION PATHS

- Special pathways in atrial wall
- Mixture of purkinje fiber and ordinary cardiac muscle cells
- Function to transmit impulses rapidly from SA node to AV node
  - **Anterior internodal pathway**
  - **Middle internodal pathway**
  - **Posterior internodal pathway**

# ANTERIOR INTERNODAL PATHWAY

- BEGINNING
  - leaves the anterior end of the sinuatrial node
- COURSE
  - passes anterior to the superior vena caval opening
  - descends on the atrial septum
- TERMINATION
  - in the atrioventricular node.

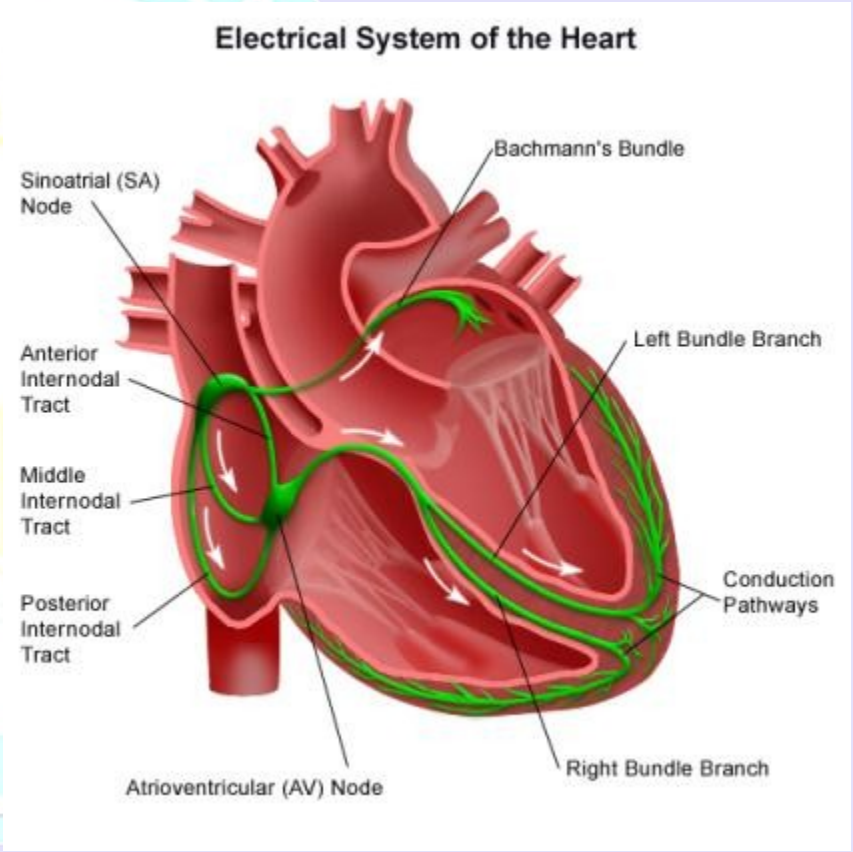


# MIDDLE INTERNODAL PATHWAY

- BEGINNING
  - leaves the posterior end of the sinuatrial node
- COURSE
  - passes posterior to the superior vena caval opening.
  - descends on the atrial septum
- TERMINATION
  - atrioventricular node.

# POSTERIOR INTERNODAL PATHWAY

- **BEGINNING**
  - Leaves the posterior part of the sinuatrial node
- **COURSE**
  - descends through the crista terminalis and the valve of the inferior vena cava
- TERMINATION
  - atrioventricular node.

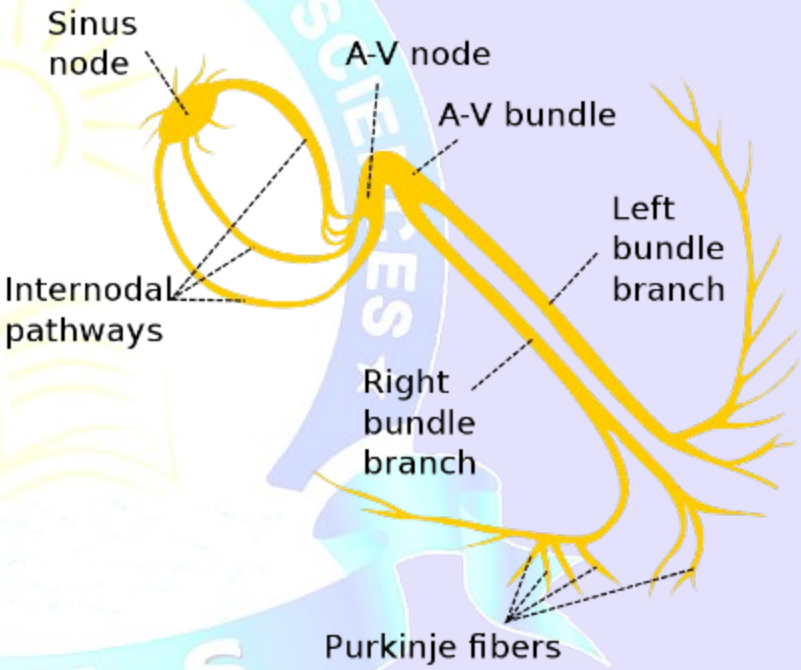


# ATRIOVENTRICULAR BUNDLE

- Bundle of His
- only pathway of cardiac muscle that connects the myocardium of the atria and the myocardium of the ventricles
- descends behind the septal cusp of the tricuspid valve
- reach the inferior border of the membranous part of the ventricular septum.
- At the upper border of the muscular part of the septum divides into two branches for each ventricle.

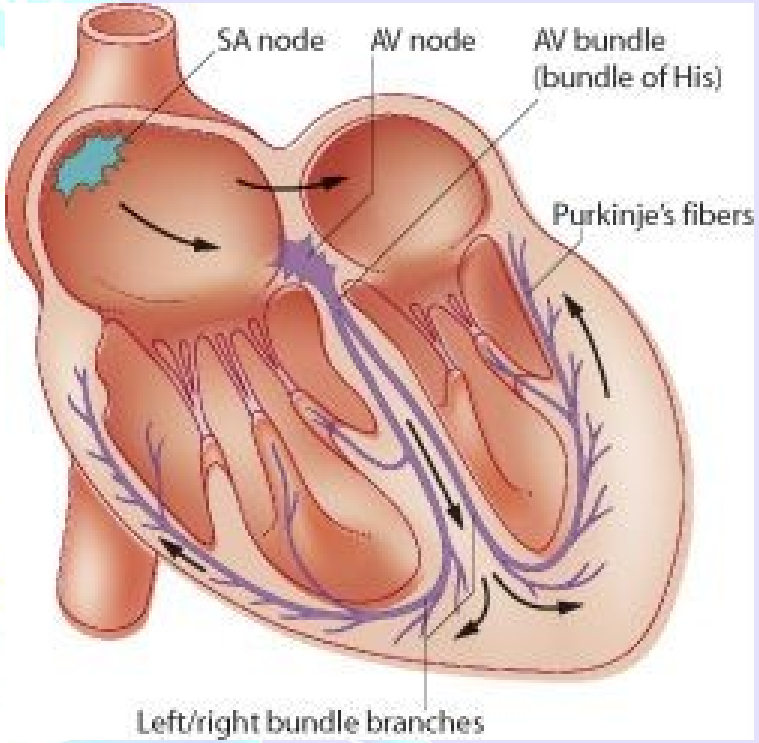
– RIGHT BUNDLE BRANCH

– LEFT BUNDLE BRANCH



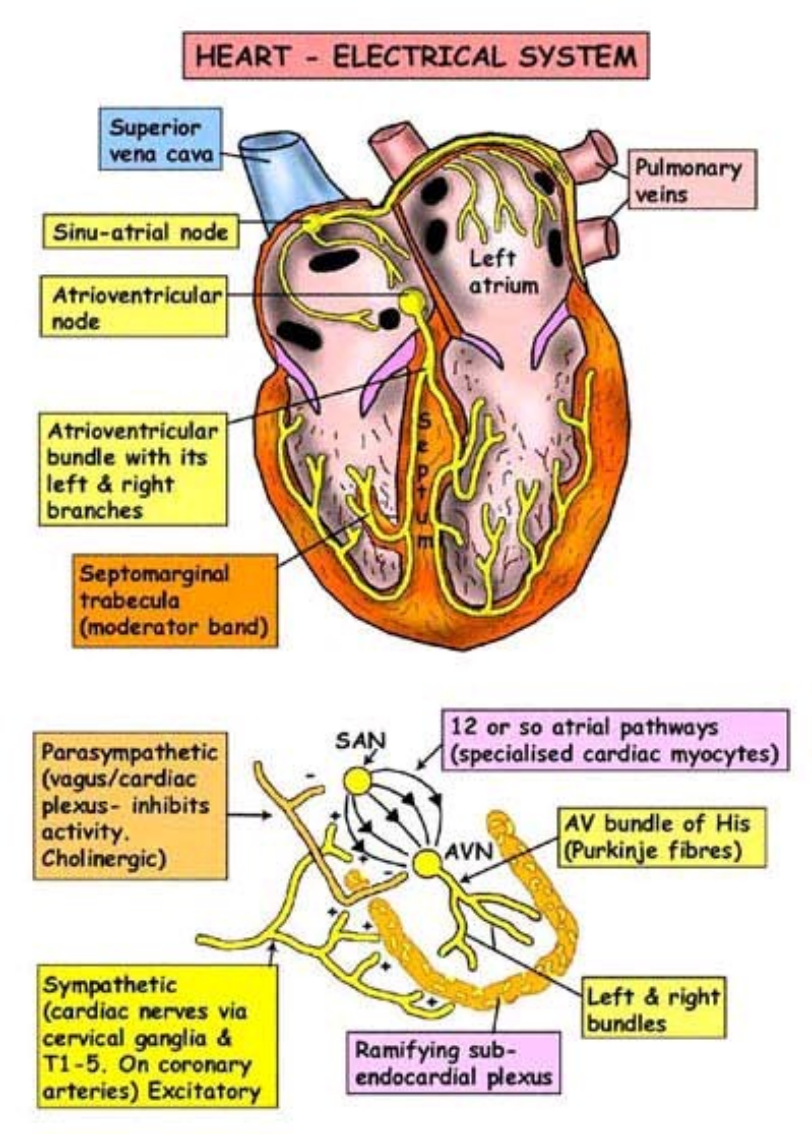
# LEFT BUNDLE BRANCH

- Pierces the **interventricular septum**
- Passes down on its left side beneath the endocardium
- Divides into two branches
  - **Anterior**
  - **Posterior**
- Eventually become continuous with the fibers of **the purkinje plexus of the left ventricle.**



# RIGHT BUNDLE BRANCH

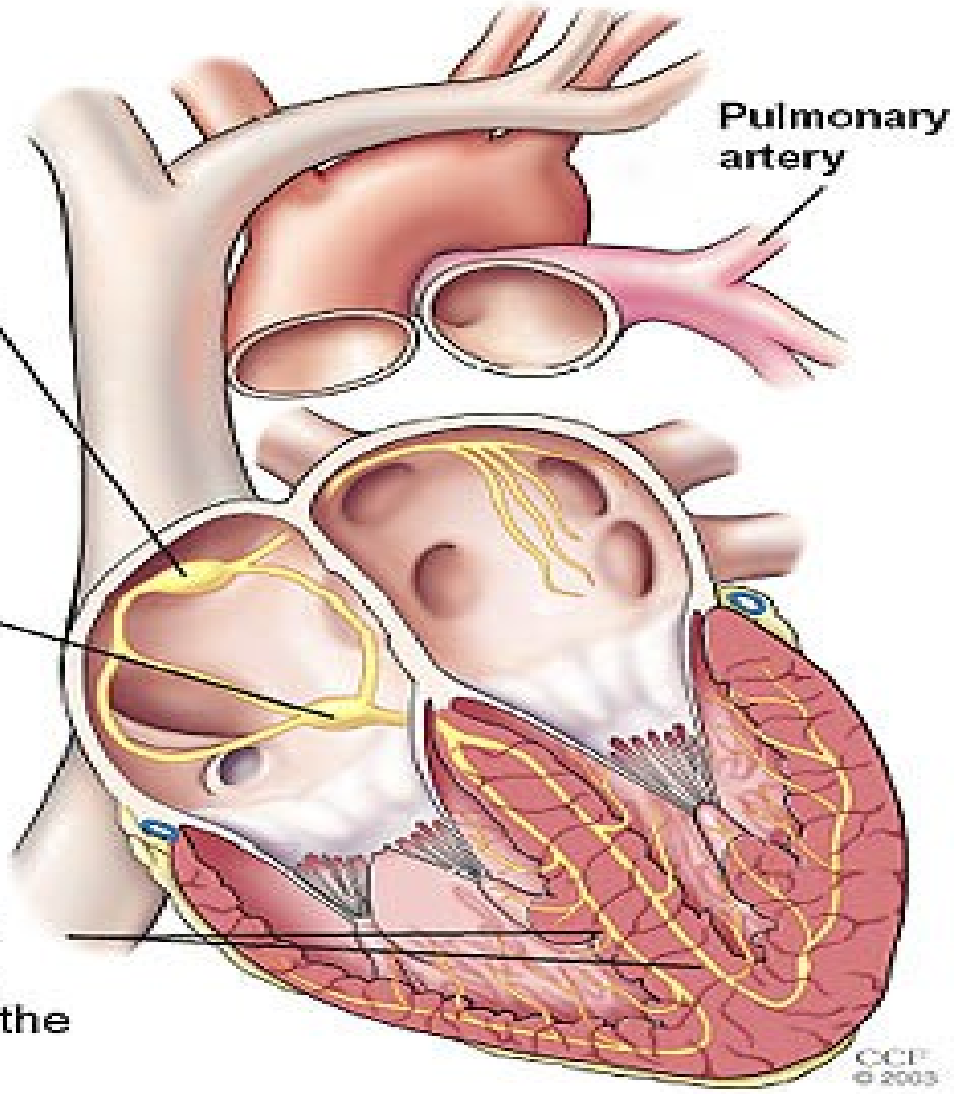
- Passes on the right side of ventricular septum
- Reach the moderatorband
- Crosses to reach anterior wall of right ventricle
- Becomes continuous with fibers of Purkinje fibers



The electrical impulse begins at the **Sinoatrial (SA) Node,** located in the right atrium. The electrical activity spreads through the walls of the atria and causes them to contract.

The **AV node** is located between the atria and ventricles and acts like a gate that slows the electrical signal before it enters the ventricles. This delay gives the atria time to contract before the ventricles.

**His-Purkinje Network**
This pathway of fibers sends the impulse into the muscular walls of the ventricles and causes them to contract.



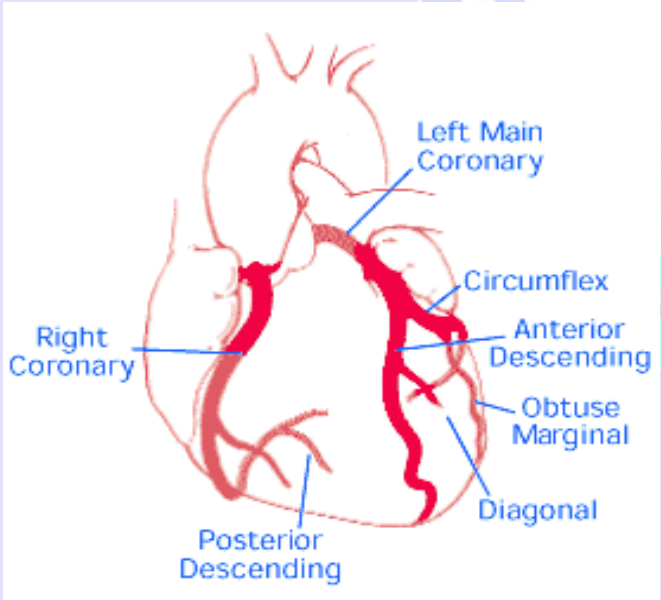
- The activities of the conducting system can be influenced by the autonomic nerve supply to the heart.

- The parasympathetic nerves slow the rhythm and diminish the rate of conduction of the impulse; the sympathetic nerves have the opposite effect.

# ARTERIAL SUPPLY OF HEART

# ARTERIAL SUPPLY OF THE HEART

- Branches of ascending aorta
  - Right coronary artery
  - Left coronary artery



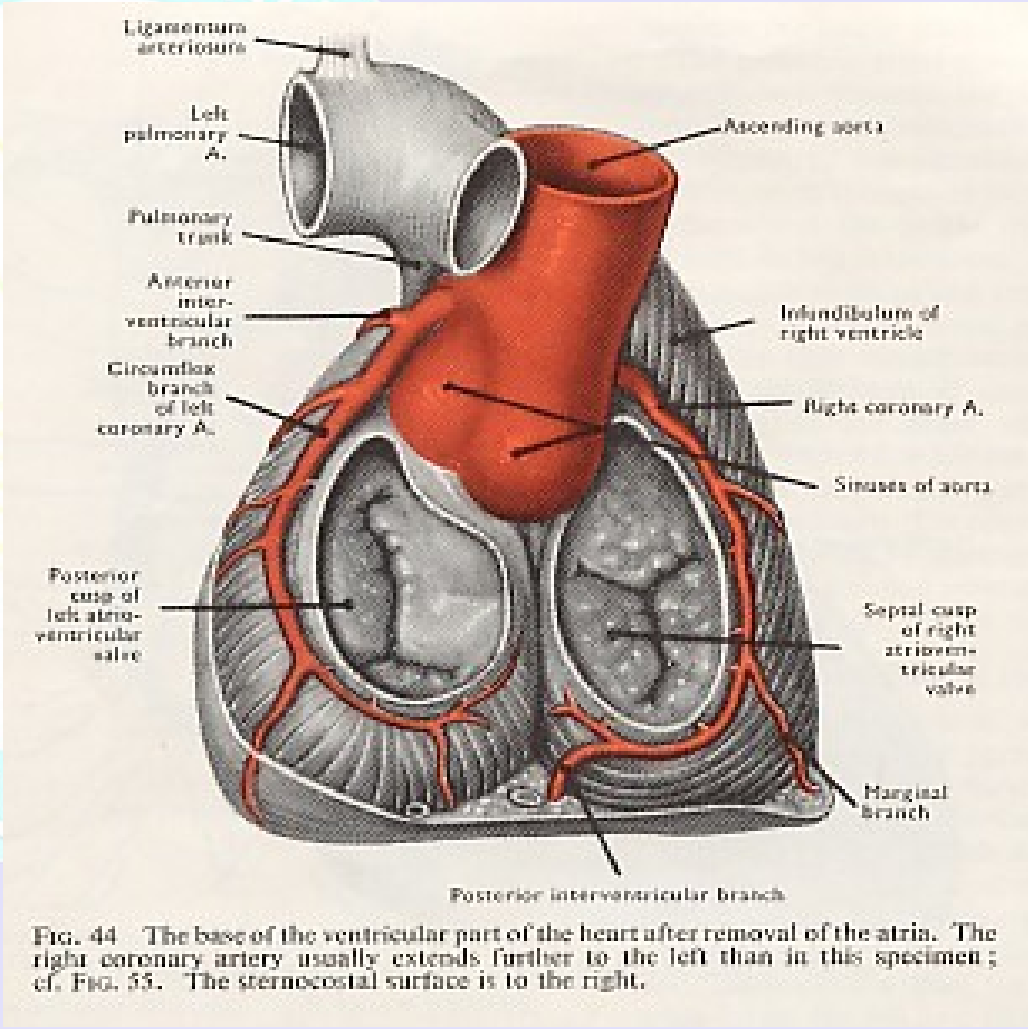




FIG. 44 The base of the ventricular part of the heart after removal of the atria. The right coronary artery usually extends further to the left than in this specimen; cf. FIG. 55. The sternocostal surface is to the right.

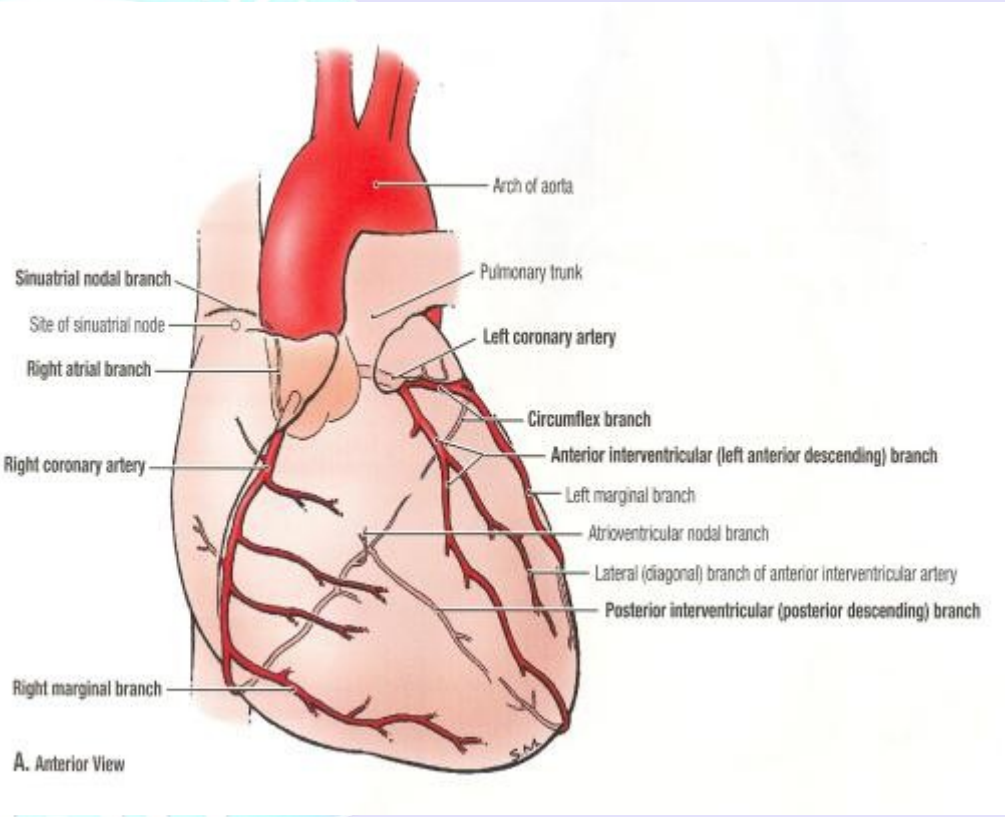
# RIGHT CORONARY ARTERY

**ORIGIN**

- Arises from the **anterior aortic sinus** of the **ascending aorta**

**COURSE**

- Runs forward between the **pulmonary trunk** and the **right auricle**
- Runs down in **right atrioventricular groove**
- At inferior border of heart,turns posteriorly
- Runs in **AV** groove
- Anastomose  with LCA in posterior interventricular groove



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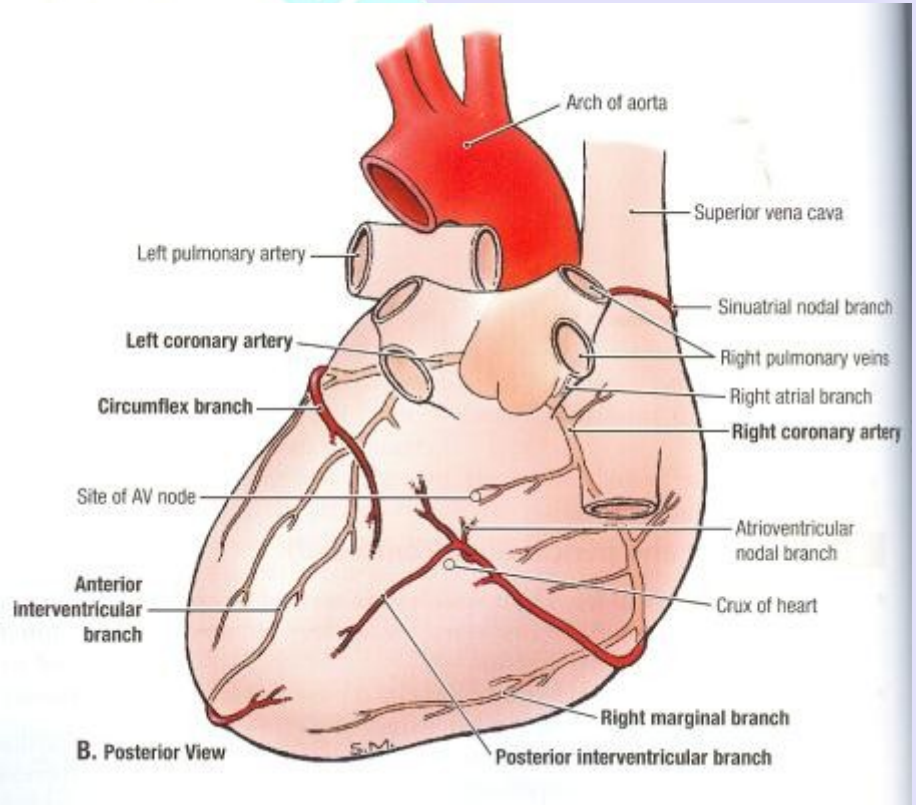
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# BRANCHES OF RIGHT CORONARY ARTERY (RCA)

- Right conus artery
- Anterior ventricular branches
  - Marginal branch
- Posterior ventricular branches
- Posterior interventricular

  ( descending artery)
- Atrial branches



# BRANCHES OF RIGHT CORONARY ARTERY RCA

**RIGHT CONUS ARTERY**

- supplies
  - **the anterior surface of the pulmonary conus (infundibulum of the right ventricle)**
  - **the upper part of the anterior wall of the right ventricle.**

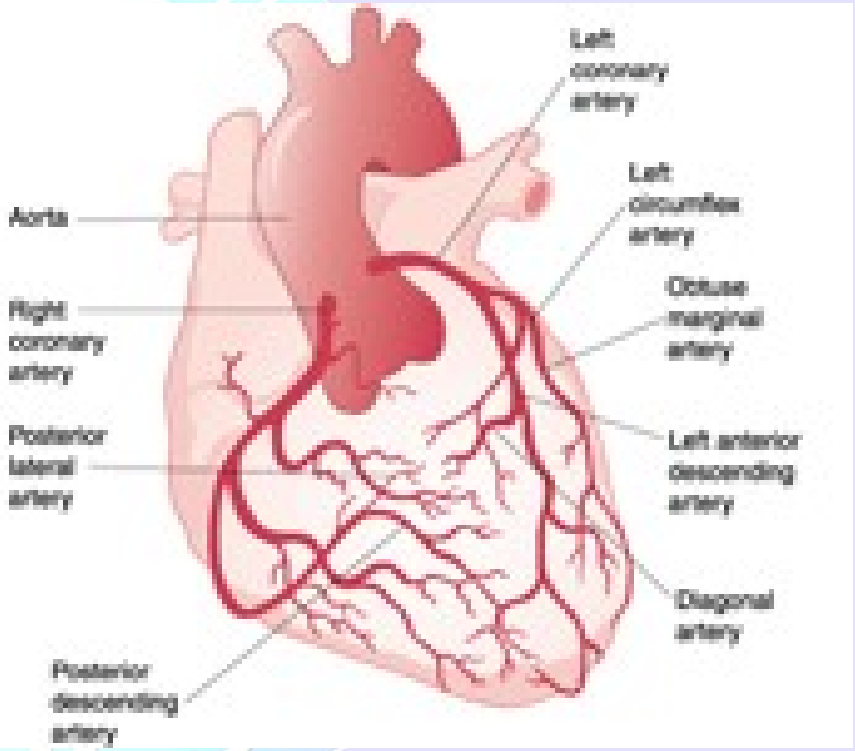
# BRANCHES OF RIGHT CORONARY ARTERY

**ANTERIOR VENTRICULAR BRANCHES**

- two or three in number
- supply the anterior surface of the right ventricle

**marginal branch**

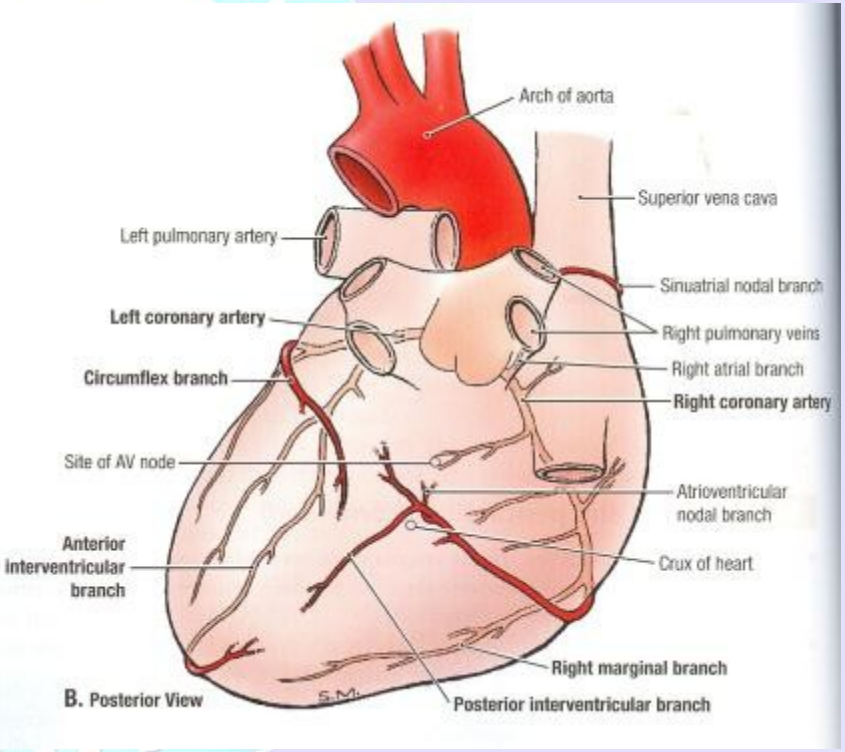
- largest and runs along the lower margin of the costal surface to reach the apex



# BRANCHES OF RIGHT CORONARY ARTERY

## POSTERIOR VENTRICULAR BRANCHES

- usually two in number
- supply the **diaphragmatic surface** of the **right ventricle**.



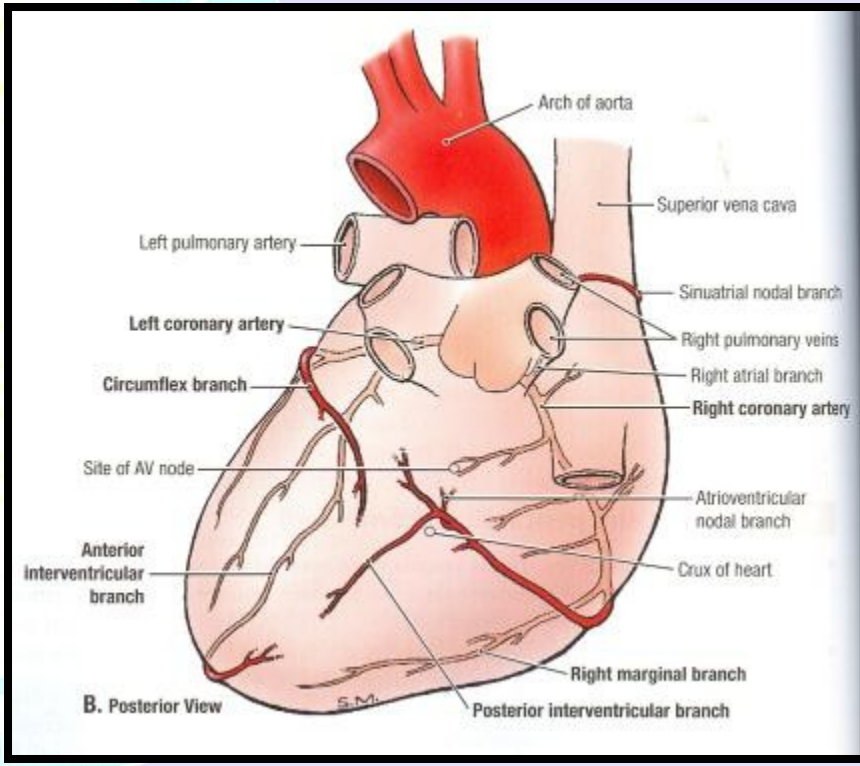
# BRANCHES OF RIGHT CORONARY ARTERY

## POSTERIOR INTERVENTRICULAR (DESCENDING) ARTERY

- runs toward the apex in the **posterior interventricular groove.**
- branches to the right and left ventricles, including its inferior wall
- Supplies **posterior part of the ventricular septum** but not to the apical part
- A large **septal branch** supplies the **atrioventricular node**.

**Variation**

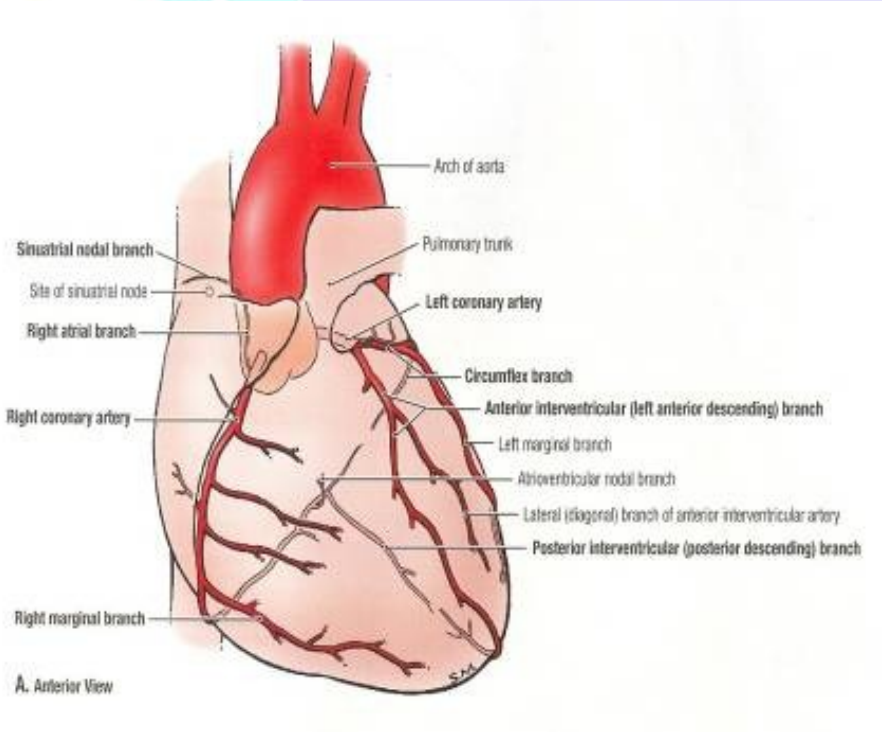
- In 10% of individuals the posterior interventricular artery is replaced by a branch from the left coronary artery



# BRANCHES OF RIGHT CORONARY ARTERY

## ATRIAL BRANCHES

- supply the anterior and lateral surfaces of the right atrium
- supplies the posterior surface of both the right and left atria.
- **artery of the sinuatrial node** supplies the node and the right and left atria;




## Variation

- in 35% of individuals it arises from the left coronary artery.

# LEFT CORONARY ARTERY (LCA)

- usually larger than the right coronary artery

**Origin**

- From left posterior aortic sinus of ascending aorta

**Course**

- Passes forward between pulmonary trunk and left auricle
- Enters inter ventricular groove

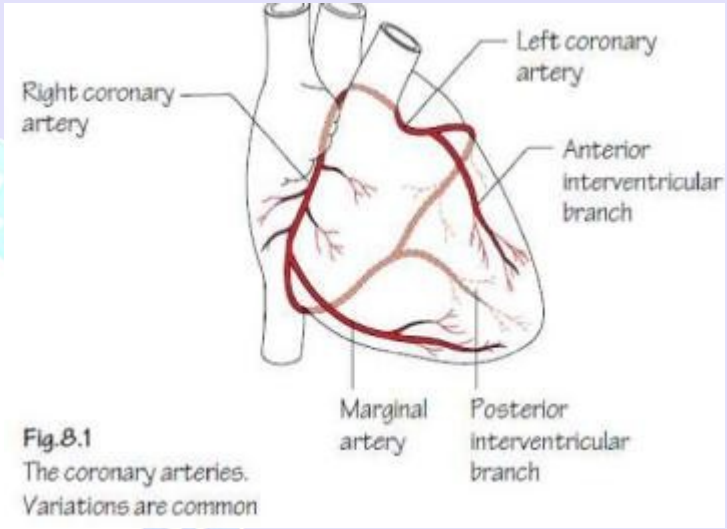



Fig.8.1
The coronary arteries.
Variations are common

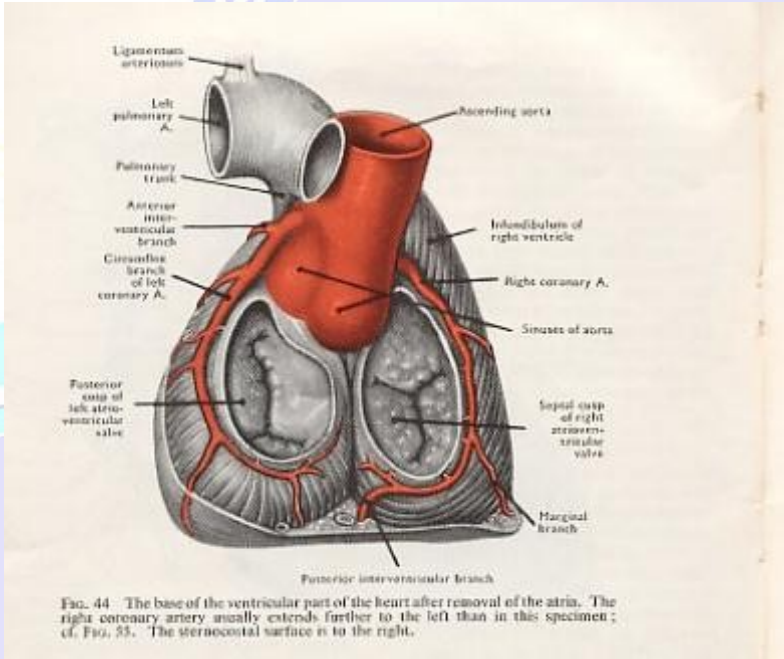
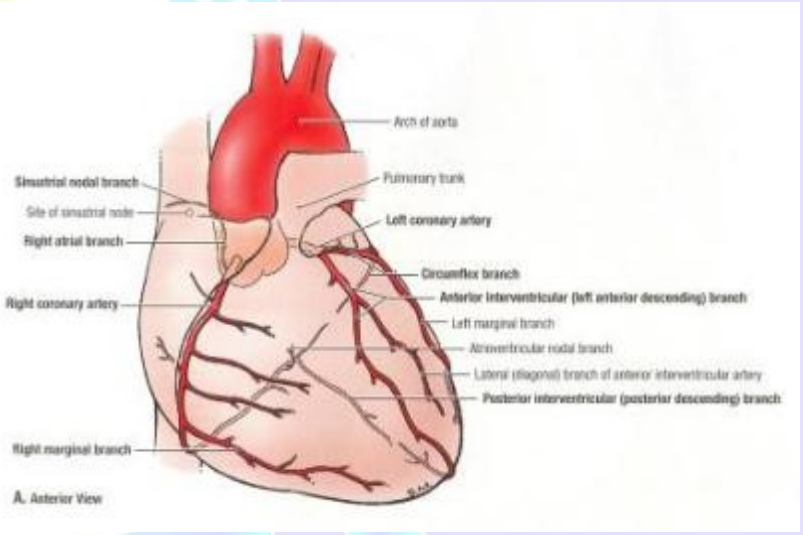



FIG. 44. The base of the ventricular part of the heart after removal of the atria. The right coronary artery usually extends further to the left than in this specimen; cf. FIG. 55. The sternocostal surface is to the right.

# BRANCHES OF LCA

## ANTERIOR INTERVENTRICULAR (DESCENDING) BRANCH

- runs downward in the anterior interventricular groove to the apex of the heart .
- supplies the right and left ventricles
- also supply the anterior part of the ventricular septum.
- One of these ventricular branches **(left diagonal artery)** may arise directly from the trunk of the left coronary artery.
- A small **left conus artery** supplies the pulmonary conus.



# BRANCHES OF LCA

## CIRCUMFLEX ARTERY

- same size as the anteriorinterventricular artery .
- winds around the left margin of the heart in the atrioventricular groove.
- A **left marginal artery** is a large branch that supplies the left margin of the left ventricle down to the apex.
- **Anterior ventricular** and **posterior ventricular branches** supply the left ventricle.
- **Atrial branches** supply the left atrium.

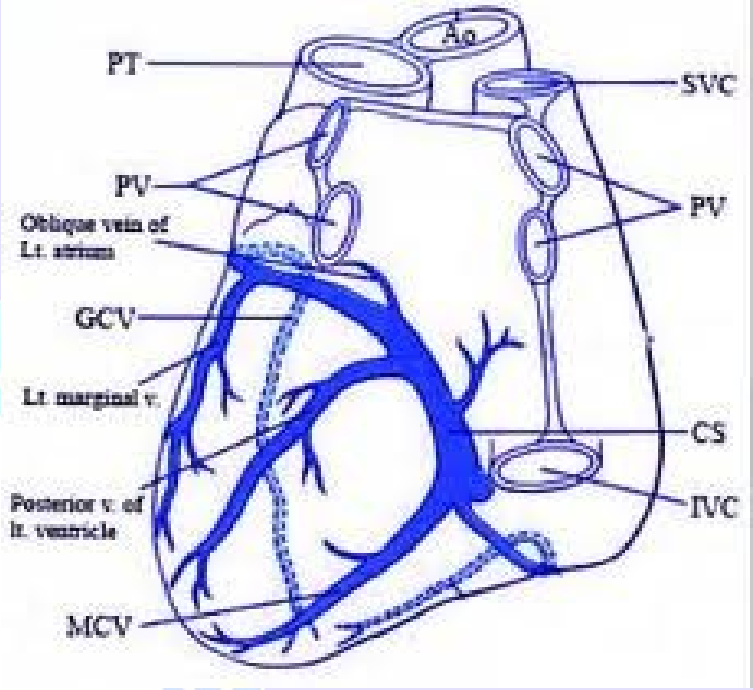
# ARTERIAL SUPPLY TO THE CONDUCTING SYSTEM

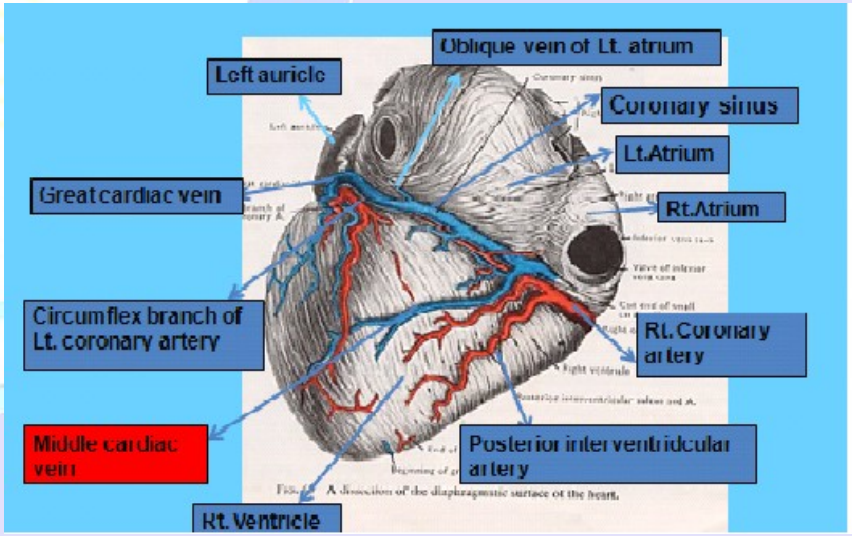
- **Sinuatrial node** by the **right**(usually)
- But sometimes by the left coronary artery.
- The **atrioventricular node** and **the atrioventricular bundle** are supplied by the **right coronary artery.**
- The **right bundle branch** of the atrioventricular bundle is supplied by the **left coronary artery;**
- The left bundle branch is supplied by the **right and left coronary arteries**

# VENOUS DRAINAGE OF THE HEART

The coronary sinus receives

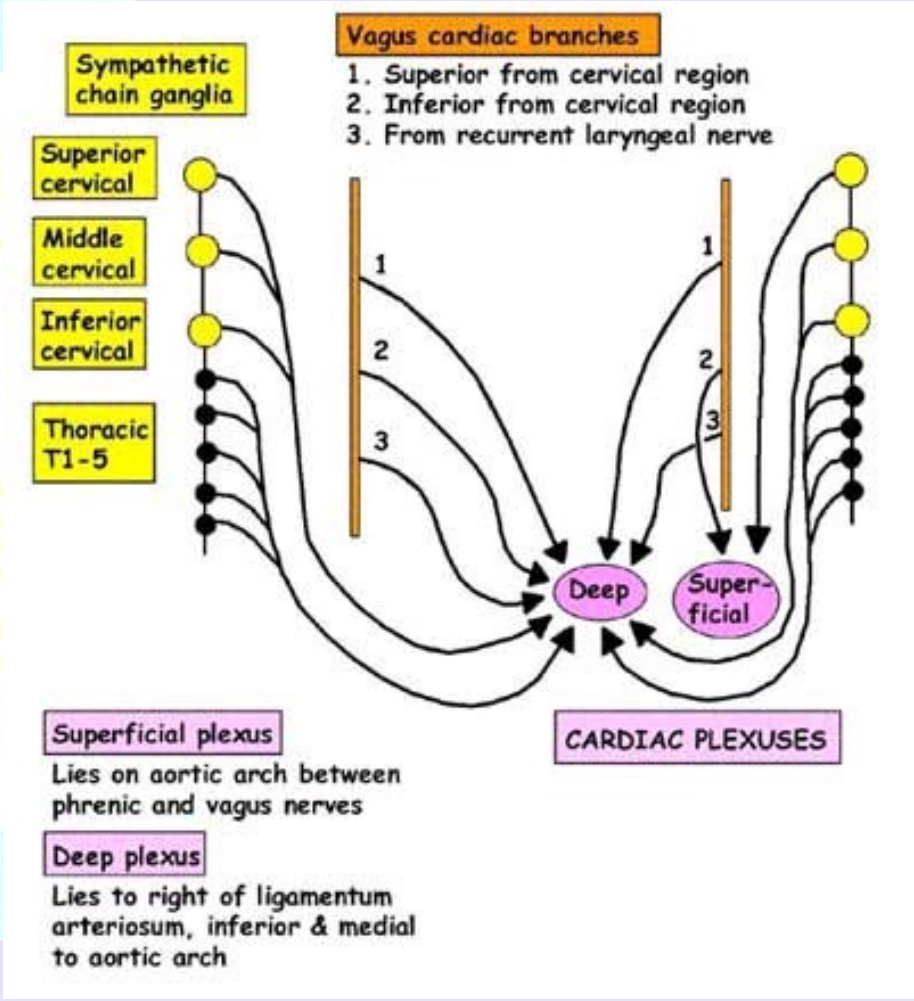
- the anterior interventricular vein or **great cardiac vein** at its **left** end
- the posterior interventricular or **middle cardiac vein**
- the **small cardiac vein** at its **right end**
- the **left posterior ventricular vein**
- The **left marginal vein** also open into the coronary sinus
- Several **small anterior cardiac vein**
- **smallest cardiac veins ( Venae cordis minimae)** are minute vessels begins in the capillary beds of the myocardium and opens directly into the chambers of the heart chiefly the atria
-



# NERVE SUPPLY OF THE HEART

- By autonomic nervous system
  - **Sympathetic fibers**
  - **Parasympathetic fibers**
- Form **cardiac plexuses** situated below the arch of the aorta
- **Sympathetic supply** from
  - **cervical and upper thoracic portions** of the sympathetic trunks
  - Increases conductivity and contractility
- **Parasympathetic supply** from
  - The **vagus** nerves.
  - Decrease conductivity and contractility






# THANK YOU